

basicity constant $K_b = \frac{\left[R\text{-}\overset{+}{N}H_3\right]\left[OH^-\right]}{\left[R\text{-}NH_2\right]}$

The basicity constant Kb gives a measure of the extent to which the amine accepts the hydrogen ion (H^+) from water,

we know that,

Larger the value of K_b or smaller the value of pK_b, stronger is the base.

Table : pK_b values of Amines in Aqueous solution. (pK_b for NH_3 is 4.74)

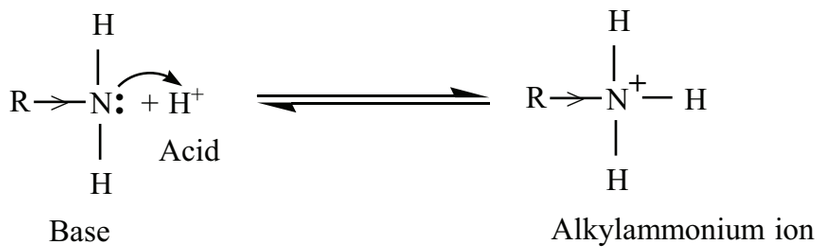
Amines	pK_b	Amines	pK_b	Amines	pK_b
$CH_3-\ddot{N}H_2$	3.38	$C_2H_5\ddot{N}H_2$	3.29	$C_6H_5CH_2-\ddot{N}H_2$	4.70
$(CH_3)_2\ddot{N}H$	3.28	$(C_2H_5)_2\ddot{N}H$	3.00	$C_6H_5-\ddot{N}HCH_3$	9.30
$(CH_3)_3\ddot{N}$	4.22	$(C_2H_5)_3\ddot{N}$	3.25	$C_6H_5\ddot{N}(CH_3)_2$	8.92

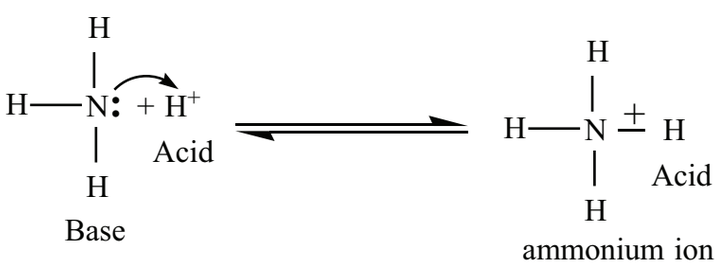
Influence of structure on basic character of amines

The factors which increase the availability of electron pair on nitrogen for sharing with an acid will increase the basic character of an amine. When a +I group like an alkyl group is attached to the nitrogen increase the electron density on nitrogen which makes the electron pair readily available for protonation.

a) Hence alkyl amines are stronger bases than ammonia.

Consider the reaction of an alkyl amine $(R\text{-}\ddot{N}H_2)$ with a proton





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